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Academic Appointments and Education

Assistant Professor	Nanjing University	2026–present	
Research Associate in Biostatistics	University of Pennsylvania	2025.06–2026	Advisor: Yong Chen
Postdoc in Biostatistics	University of Pennsylvania	2023.08–2025.06	Advisor: Yong Chen
Ph.D. in Statistics	Peking University	2018.09–2023.06	Advisor: Wei Lin
B.S. in Statistics	Nankai University	2014.09–2018.06	

Publications and Preprints

* denotes equal contribution (co-first authors).

- **Published**

1. Xiang Li, Feng Ruan, **Huiyuan Wang**, Qi Long, and Weijie J. Su. Robust Detection of Watermarks for Large Language Models under Human Edits. *Journal of the Royal Statistical Society Series B: Statistical Methodology* 88(2), 2026, 491–515.

2. Dazheng Zhang*, Bingyu Zhang*, **Huiyuan Wang*** **(co-first authors)**, Yiwen Lu, Charles J. Wolock, Wenjie Hu, Linbo Wang, George Hripcsak, and Yong Chen. Negative-Control-Calibrated Difference-in-Difference Analyses: Addressing Unmeasured Confounding in RWD with Application to Racial/Ethnic Differences. *npj Digital Medicine* 8(1), 2025, Article 452.

3. Xiang Li, Feng Ruan, **Huiyuan Wang**, Qi Long, and Weijie J. Su. A Statistical Framework of Watermarks for Large Language Models: Pivot, Detection Efficiency and Optimal Rules. *The Annals of Statistics* 53(1), 2025, 322–351.

4. Shucong Zhang, **Huiyuan Wang**, and Wei Lin. CARE: Large Precision Matrix Estimation for Compositional Data. *Journal of the American Statistical Association: Theory and Methods* 120(549), 2025, 305–317.

5. Xuyang Zhao, **Huiyuan Wang**, Weiran Huang, and Wei Lin. A Statistical Theory of Regularization-Based Continual Learning. In *Proceedings of the 41st International Conference on Machine Learning (ICML)*, PMLR 235, 2024, 61021–61039.

6. Caizhi Tang*, **Huiyuan Wang*** **(co-first authors)**, Xinyu Li, Qing Cui, Longfei Li, and Jun Zhou. Difference-in-Differences Meets Tree-based Methods: Heterogeneous Treatment Effects Estimation with Unmeasured Confounding. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*, PMLR 202, 2023, 33792–33803.

7. Xuyang Zhao, **Huiyuan Wang**, and Wei Lin. The Aggregation–Heterogeneity Trade-off in Federated Learning. In *Proceedings of the 36th Conference on Learning Theory (COLT)*, PMLR 195, 2023, 5478–5502.

8. Caizhi Tang*, **Huiyuan Wang*** **(co-first authors)**, Xinyu Li, Qing Cui, Ya-Lin Zhang, Feng Zhu, Longfei Li, Jun Zhou, and Linbo Jiang. Debiased Causal Tree: Heterogeneous Treatment Effects Estimation with Unmeasured Confounding. *Advances in Neural Information Processing Systems* 35, 2022, 5628–5640.

- **Accepted**

9. Jingyue Huang, Yuqing Lei, **Huiyuan Wang**, and Yong Chen. ANCHOR: Double Machine Learning for NCO-Calibrated Adaptive Marginal Sensitivity Bounds. *The Annals of Applied Statistics*. Accepted for publication on October 2, 2026.

10. Yong Chen, Yiwen Lu, Jingmei Qiu, **Huiyuan Wang**, and Yudong Wang. MOSAiC: Multi-site One-Shot Aggregation of Compressed Risk Functions. *Journal of the American Statistical Association: Applications and Case Studies (with Discussion)*, 2026, pp. 1–15 (advance online publication). DOI: 10.1080/01621459.2026.2698025. (In Alphabetical Order)

- **Under Revision**

11. Jianmin Chen, **Huiyuan Wang**, Thomas Lumley, Xiaowu Dai, and Yong Chen. Surrogate-Powered Inference: Regularization and Adaptivity. *arXiv:2512.21826*. *Journal of the American Statistical Association: Applications and Case Studies (Moderate Revision)*, 2026+.

12. Yuru Zhu, **Huiyuan Wang**, Haitao Chu, Yumou Qiu, and Yong Chen. Collaborative Indirect Treatment Comparisons with Multiple Distributed Single-Arm Trials. *arXiv:2509.23664*. *Journal of the American Statistical Association: Theory and Methods (Revise and Resubmit)*, 2026+.

13. **Huiyuan Wang** and Wei Lin. Nonasymptotic Theory for Two-Layer Neural Networks: Beyond the Bias–Variance Trade-off. *arXiv:2106.04795*. *Journal of Machine Learning Research (Revise and Resubmit)*, 2026+.

- **Manuscript**

14. Jingyue Huang, **Huiyuan Wang**, Yuqing Lei, and Yong Chen. Efficient Semiparametric Inference for Distributed Data with Blockwise Missingness. *arXiv:2508.16902*.

15. Yalong Lyu*, **Huiyuan Wang*** (co-first authors), and Wei Lin. Temporal Point Process Graphical Models. *arXiv:2110.11562*.

16. **Huiyuan Wang**, Xuyang Zhao, and Wei Lin. Heterogeneous Federated Learning on a Graph. *arXiv:2209.08737*.

Awards

1. Top Reviewer at NeurIPS 2026, 2025
2. The Excellent Poster Award of the The Sixth Peking–Tsinghua University Statistical Forum 2022
3. The Second Prize of the Fourth National Academic Forum for Doctoral Students in Statistics (Chinese Association for Applied Statistics) 2020
4. Outstanding Graduate (Nankai University) 2018

Teaching Experience

1. Recitation Instructor for *Advanced Mathematics B*, School of Mathematical Sciences, Peking University, Fall 2022
2. Teaching assistant for *Advanced Theory of Statistics II*, School of Mathematical Sciences, Peking University, Spring 2021
3. Recitation Instructor for *Advanced Mathematics D*, School of Mathematical Sciences, Peking University, Fall 2020
4. Teaching assistant for *Modern Statistical Modeling*, School of Mathematical Sciences, Peking University, Spring 2020
5. Teaching assistant for *Statistical Learning*, School of Mathematical Sciences, Peking University, Fall 2019

Talks

1. *Calibrated Digital Twins: Improving RCT Analysis with Distributionally Shifted RWD* at an *Invited Session* on Joint Statistical Meetings, Nashville, 2025
2. *Augmented Targeted Trial Emulation* on Joint Statistical Meetings, Nashville, 2025
3. *Negative Control-Calibrated Difference-in-Difference Analyses: Addressing Unmeasured Confounding in Real-World Data with Application to Quantifying the Impact of Pandemic on Racial/Ethnic Differences* on Joint Statistical Meetings, Nashville, 2025
4. *Augmented Targeted Trial Emulation* on International Chinese Statistical Association Applied Statistics Symposium, Storrs, 2025
5. *Efficient and robust high-dimensional hypothesis testing with surrogate outcomes* on International Chinese Statistical Association Applied Statistics Symposium, Nashville, 2024

6. *CARE: Large precision matrix estimation for compositional data* on The Fourth National Academic Forum for Doctoral Students in Statistics (held by Chinese Association for Applied Statistics), Guangzhou, 2020

Professional Service

1. *Program Committee Member* for Algorithmic Learning Theory (ALT), 2027
2. *Area Chair* for ICLR, 2026
3. *Reviewer* for NeurIPS, 2026, 2025
4. *Reviewer* for ICML, 2026
5. *Reviewer* for Algorithmic Learning Theory (ALT), 2023–2026
6. *Reviewer* for The Annals of Applied Statistics (AOAS), 2026
7. *Reviewer* for JASA, 2026, 2025, 2023
8. *Reviewer* for ICLR, 2025
9. *Reviewer* for the *Best Student Paper Award*, International Chinese Statistical Association, 2025
10. *Organizer* of an invited session for International Chinese Statistical Association, 2025
11. *Reviewer* for JCGS, 2022

Memberships

1. American Statistical Association
2. International Chinese Statistical Association
3. The Society for Causal Inference
4. Causal Inference Branch of the Chinese Association for Applied Statistics